AI-Powered Supply Chain Demand Forecasting

Detailed explanation of every supplied dashboard image

Each submitted visual crop and each full-page dashboard screenshot is explained individually. Displayed values are transcribed from the screenshots; definitions and calculations should be validated against the Power BI model and source data.

Page 1 — Executive Overview

Image-by-image guide

KPI: Total Units Sold — 9.98M



Shows the total quantity of units sold in the selected date range. It is the primary demand-volume measure on this page. Compare it with ordered units and inventory, while confirming the exact aggregation and units in the data model.

Key Insights strip



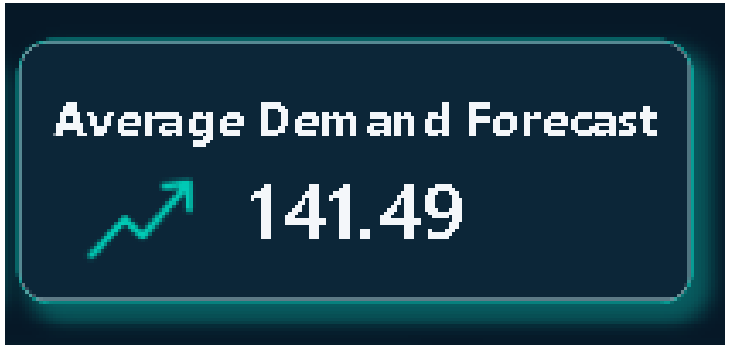
Summarizes the dashboard’s highlighted messages: +4.7% units sold versus the previous period, Electronics as the highest category (20.3%; 2.03M units), East region at 2.51M, and inventory of 10.57M (+5.1%). These are displayed callouts; reconcile their definitions and filter context with the KPI cards before treating them as validated results.

KPI: Total Units Ordered — 8.04M



Represents total ordered units. Comparing it with units sold can help assess the relationship between order activity and sales, but it does not by itself establish whether orders were fulfilled or backlogged.

KPI: Average Demand Forecast — 141.49



Displays the average forecast measure. The screenshot does not specify the forecast grain (for example, per day, product, or record), so check the measure definition before interpreting the magnitude.

KPI: Inventory Level — 20M



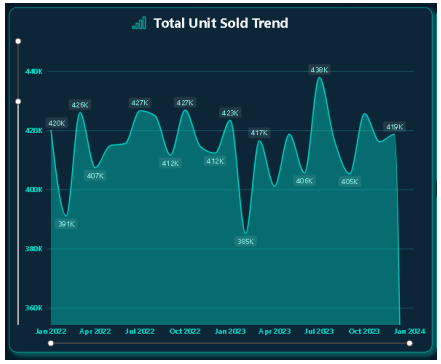
Displays the inventory quantity. Interpret alongside sales and order volume; inventory may represent a snapshot or aggregated records, so the measure’s time grain matters.

Date Range slicer



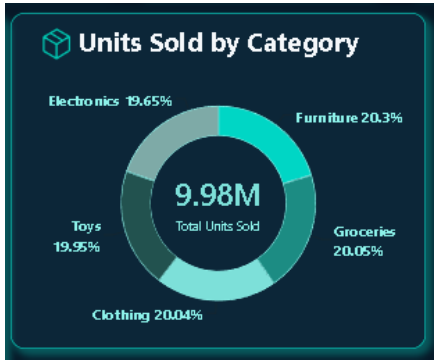
Shows the report's date window as 01 Jan 2022–01 Jan 2024. It is intended to filter the visuals to a selected period. Test that all visuals respond and that previous-period calculations use the correct comparison window.

Chart: Total Unit Sold Trend



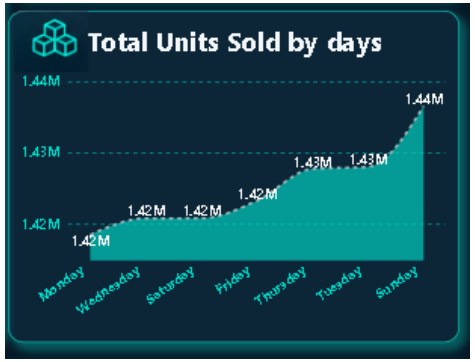
A time-series chart showing sales fluctuations over the report period. Labeled values vary roughly from 385K to 438K in the visible series. Use it to identify peaks and dips; it describes movement, not the cause of that movement.

Chart: Units Sold by Category



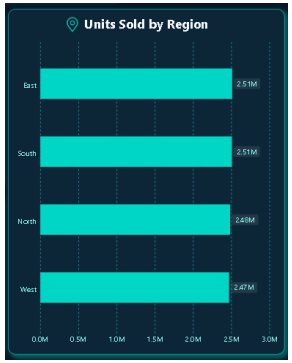
A donut chart showing category shares of the 9.98M total: Furniture 20.3%, Groceries 20.05%, Clothing 20.04%, Toys 19.95%, Electronics 19.65%. The categories are relatively evenly distributed, with no large share gap.

Chart: Total Units Sold by Days



Compares units sold by weekday. The displayed totals rise from around 1.42M early in the week toward about 1.44M on Sunday. Validate the weekday aggregation and exact values before drawing operational conclusions.

Chart: Units Sold by Region



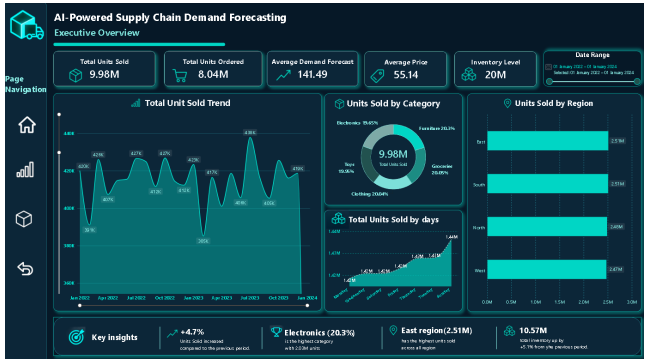
Horizontal bars compare East (2.51M), South (2.51M), North (about 2.48M), and West (about 2.47M). Regional volumes are close; East and South appear slightly higher in the displayed report.

Page heading: Executive Overview



The title banner identifies the report and the current page. The subtitle sets the context for the KPI and chart group that follows.

Full dashboard: Executive Overview



The main page combines headline sales and inventory KPIs, date filtering, the sales trend, category and weekday breakdowns, regional comparison, and the insight strip. Read the cards for totals, then use the charts to explore when, where, and which categories contribute to the total.

Page 2 — Inventory & Operations

Image-by-image guide

KPI: Total Inventory — 20.1M



Displays total inventory. Confirm whether this is a stock snapshot, an average, or a sum across rows/dates; summing inventory snapshots over time can misrepresent stock on hand.

KPI: Low Inventory Records — 8.0K



Counts records classified as low inventory. The low-stock threshold and the record grain are not visible here and should be defined in the report documentation.

KPI: Average Inventory Level — 274.47



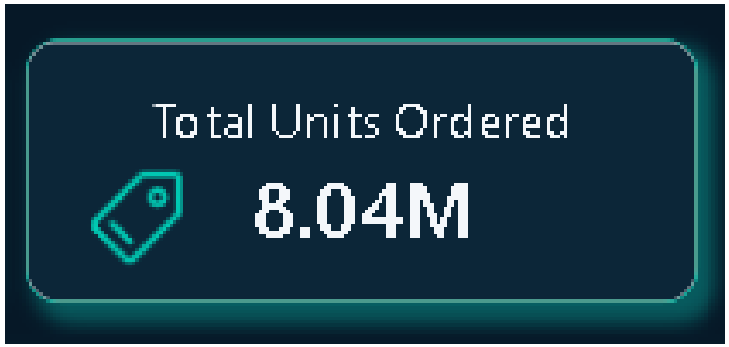
Shows the average inventory value. Establish whether the average is per product, store, date, or source row before comparing it with the total inventory KPI.

KPI: Total Units Ordered — 8.04M



Summarizes order quantity for the selected period. Compare with inventory and sales to examine replenishment and demand context.

KPI: Average Discount — 10.01%



Displays average discount percentage. Confirm if this is a simple row average or weighted by quantity/revenue; those methods can produce different results.

Date Range slicer



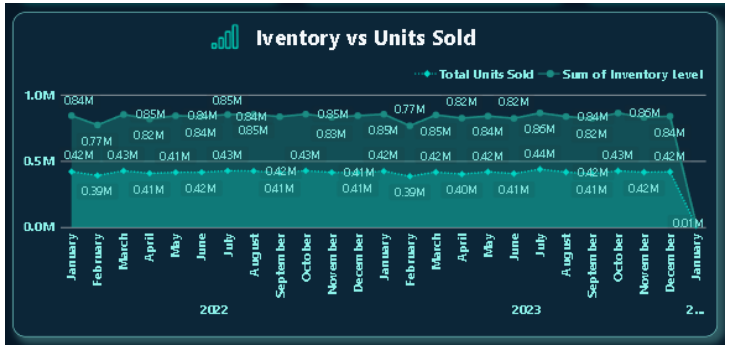
Sets the date interval used by the inventory and operations visuals. Validate how the selection affects inventory snapshots and date-based totals.

Chart: Inventory vs Units Sold



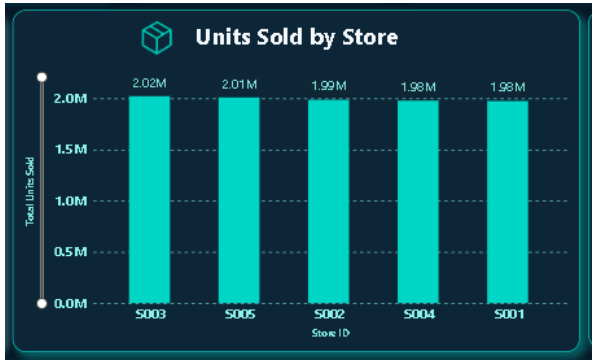
Compares inventory level and units sold across months. Look for periods where the two measures diverge. Make sure the measures use compatible units and valid time aggregations before comparing their heights.

Chart: Units Sold by Store



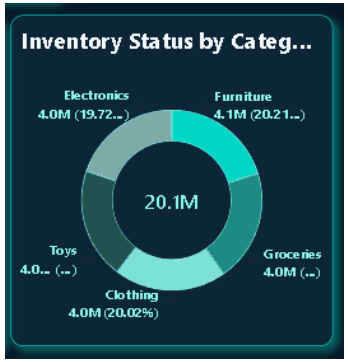
Ranks store sales in the shown view: S003 2.02M, S005 2.01M, S002 1.99M, S004 1.98M, and S001 1.98M. The values are close, suggesting similar total sales in this view.

Chart: Inventory Status by Category



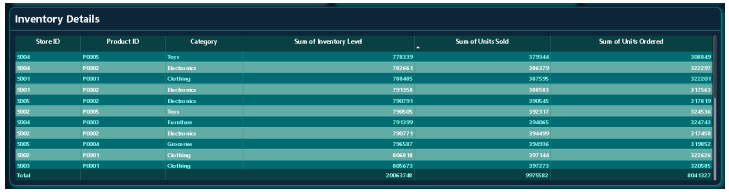
A donut chart showing inventory split across product categories, with displayed amounts around 4.0M–4.1M. Some labels are truncated, so use the underlying visual tooltip or data table for exact category amounts.

Table: Inventory Details



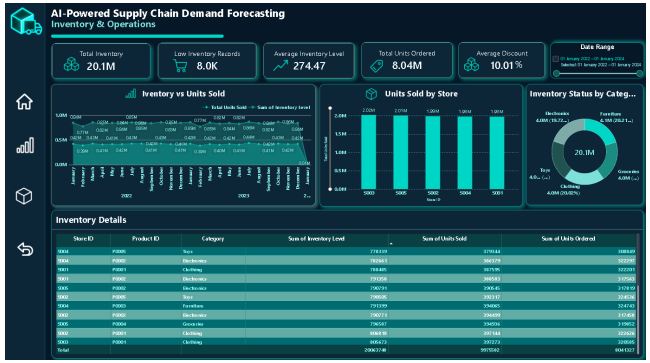
Provides detail by Store ID, Product ID, and Category, alongside inventory level, units sold, and units ordered. Use it to inspect individual store-product combinations after spotting an issue in an aggregate chart.

Page heading: Inventory & Operations



Identifies the page's operational focus: inventory, ordering, discounts, and store/category stock patterns.

Full dashboard: Inventory & Operations



The complete page brings together inventory and order KPIs, a date slicer, monthly inventory-versus-sales comparison, store sales, category inventory composition, and a detailed inventory table.

Page 3 — Demand Forecast & Analysis

Image-by-image guide

KPI: MAE — 135.39



Mean Absolute Error is the average absolute difference between actual and predicted units. It is expressed in the target's units; lower values indicate smaller average errors when comparing the same test set.

KPI: RMSE — 176.16



Root Mean Squared Error penalizes larger errors more strongly than MAE. It is also in the target's units and is most useful when comparing models on the same data.

KPI: R² Score — 0.71



R-squared indicates how much variation in the evaluated target is accounted for by the model relative to a baseline. It is not a 71% accuracy score.

KPI: MAPE — 0.45



Mean Absolute Percentage Error is displayed as 0.45. Confirm formatting: this could represent a decimal ratio (45%) or a percentage-formatted value. The screenshot alone does not establish the intended interpretation.

KPI: Test Records — 3K



Shows an approximate test-set record count. Confirm the exact number and ensure the evaluation metrics are calculated over the same test records.

Date Range slicer



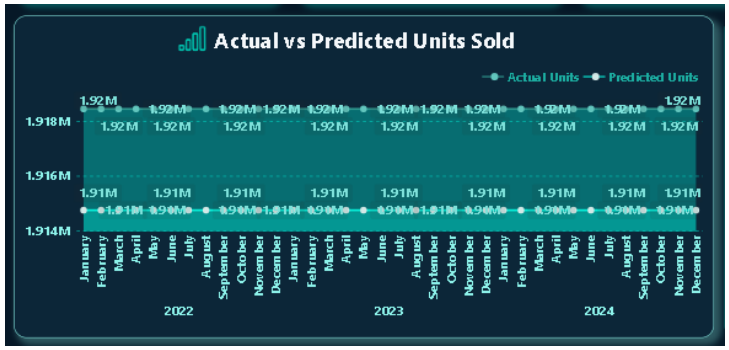
Displays the 01 Jan 2022–01 Jan 2024 range. For model evaluation, ensure the slicer does not mix training-period records into metrics intended to represent test-set performance.

Chart: Actual vs Predicted Units Sold



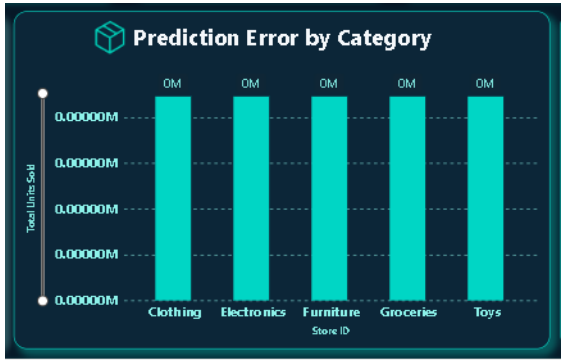
Plots actual and predicted units over time. Close lines can suggest similar overall movement, but exact residuals are needed to judge error; visual proximity alone is not a full accuracy assessment.

Chart: Prediction Error by Category



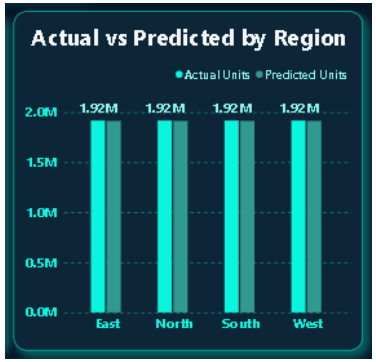
Compares prediction error across Clothing, Electronics, Furniture, Groceries, and Toys. The displayed labels appear near zero and the scale is difficult to read; check the error measure, formatting, and aggregation before interpreting.

Chart: Actual vs Predicted by Region



Compares actual and predicted units for East, North, South, and West. Bars appear similar at the displayed scale. Use exact values or a difference measure to identify regional over- or under-prediction.

Table: Highest Prediction Errors



Lists year, quarter, month, day, store, product, actual units, predicted units, and absolute error. It is intended to surface records with the largest misses. The visible rows repeat identical values, so check sorting, row-level grain, relationships, and whether aggregated values are being repeated.

Page heading: Demand Forecast & Analysis



Identifies the report page devoted to model evaluation and actual-versus-predicted comparisons.

Full dashboard: Demand Forecast & Analysis



Combines error metrics, date filtering, actual-versus-predicted trend, category error, regional comparison, and a detailed error table. Start with the metrics for overall evaluation, then use visuals and the table to locate where errors occur.